

Figure 4: Todo WebApp on Opera Mobile browser

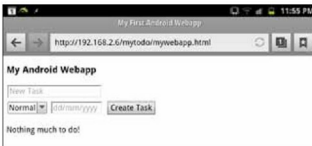


Figure 5: Todo WebApp on Android default browser

the app does not have Internet connectivity. Native apps don't have this limitation.

- **Ease of use:** A native app may score higher on this count as it runs as an independent application and it's easy to switch between multiple applications on an Android device. An application with an icon for the App is more user-friendly when it is being launched.
- **User satisfaction ratings and reviews:** Users like to review apps and compare them before installation. This is more convenient when it is a native app.
- **Monetising:** Native apps score better at monetising, due to the transaction infrastructure already provided by Google.
- **Responsiveness:** The main difference between WebApps and native apps is that native apps have the UI already on the client, and have the luxury of choosing rich graphics that are stored locally. WebApp developers need to be conscious of the download time and data transfer limits.
- **Hardware performance:** While WebApps make it easier to use the hardware sensors available on Android mobiles, their effectiveness and performance is better with native apps.

WebApp support in Android

Android supports native apps and WebApps in the browser.

It is also possible to develop hybrid apps -- native apps with a WebApp embedded, using the WebView feature on Android, supported from O.S. Version 1.0.

Why use Web technologies?

The World Wide Web has come a long way. Though there is standardisation of Web technologies such as HTML, different browser vendors are divided with respect to support. Some vendors have introduced proprietary features in browsers, leaving many websites incompatible, or inconveniencing programmers because of the need for additional coding and testing. The most common banner on websites, for a while, was, "Works only in IE". This was the situation with desktop-based browsers.

The Android browser uses WebKit as its rendering engine, which is close to W3C (World Wide Web Consortium) standards, and is making Open Web adoption a reality. An appeal from the W3C, regarding the HTML5 series of technologies, is to eliminate incompatibilities between browsers. With WebKit-based browsers compatible across Android and iOS-based devices, the majority of the mobile Web is closer to standards than earlier, benefiting both developers and users.

The HTML5 series of technologies is a big leap forward in standardisation, and the W3C acknowledges that the current shipping versions of all major browsers have good support for key HTML5 features. Mobile gaming, major news websites and financial services have started adopting the latest features, for a better user experience. HTML5, in combination with JavaScript and CSS3, brings the mobile Web experience close to native apps.

Your first WebApp

Developing and testing your first Web app is no different from a Web page. You can use a text editor (like vi or Notepad) to enter the following HTML code, and save it as "myfirstwebapp.html". Copy the HTML file to your Android mobile, and open it in the browser. You could also test your WebApp on a wireless LAN network by saving the HTML file on a HTTP server (such as XAMPP) and entering the IP address (or hostname) as the URL in the browser:

```
<!doctype html>
<html>
<head>
<style type="text/css">
p.sh
{
    border-style: solid;
    border-width: medium;
    -webkit-border-radius: 15px;
    background-color: yellow;
}
</style>
```